COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 / Data Structures and Algorithms	**SECTION:**	CPE21S3
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ACTIVITY NO. #4 **Array-Based vs STL Stack Implementations**

1. PROCEDURE OUTPUT

ILO A:

```cpp
#include <iostream>
#include <stack> // call the stack from the STL

int main(){

    //testing
    std::stack<int> newStack; // created an object
    //testing the empty() function to check if the stack is empty
    std::cout<<"Is the stack empty? "<< newStack.empty() << std::endl;

    //insert 3 elements:
    newStack.push(3);
    std::cout << "Top of the stack: " << newStack.top() << std::endl;
    newStack.push(2);
    std::cout << "Top of the stack: " << newStack.top() << std::endl;
    newStack.push(1);
    std::cout << "Top of the stack: " << newStack.top() << std::endl;

    std::cout << "Is the stack empty? " << newStack.empty() << std::endl;
    std::cout << "Size of the stack: " << newStack.size() << std::endl;

    std::cout << "Is the stack empty? " << newStack.empty() << std::endl;
    newStack.push(6);
    newStack.push(7);
    newStack.push(8);

    return 0;
}
```

Output of the Code:

```
Is the stack empty? 1
Top of the stack: 3
Top of the stack: 2
Top of the stack: 1
Is the stack empty? 0
Size of the stack: 3
Is the stack empty? 0
```

Analysis:

Asides from the usual header file, the code used another header file which is #include <stack> this way it will call the stack from the STL. In the main function, we used std::stack<int> newStack to create an object. To see if it's working or if it's empty, we print it via cout and it shows 1 as a boolean meaning that it is empty. We also insert 3 elements that we pushed, each appearing based on their positions in the code. We also asked if the stack was empty and the size of the stack, it answered as 0 meaning its not true and its false, it answered 3 since the stack is 3. We also include an additional stack and print if the stack is empty which says 0 meaning it's false.

ILO B:

ILOA > main.cpp > ...

```cpp
#include <iostream>

const size_t maxCap = 100;
int stack[maxCap];
int top = -1;
int newData, elements, choice;

//prototyping

bool isEmpty();
bool isFull();
void push();
void pop();
void Top();
//displayALL
void displayAll();

int main(){
    std::cout<< "Enter the number of maximum elements for the huge stack (Max is 100): ";
    std::cin >> elements;

    while(true){
        std::cout << "Stack Operations: " << std::endl;
        std::cout << "1. Push\n2. Pop\n3. Top\n4. isEmpty\n5. isFull\n6. Display All" << std::endl;
        std::cout << "Enter a number: ";
        std::cin >> choice;

        switch(choice){
            case 1: push();
            break;
            case 2: pop();
            break;
            case 3: Top();
            break;
            case 4: std::cout << isEmpty() << std::endl;
            break;
            case 5: std::cout << isFull() << std::endl;
            break;
            case 6: displayAll();
            break;
            default: std::cout << "Invalid choice. " << std::endl;
            break;
        }
    }
    return 0;
}
bool isEmpty(){
    //top is -1
```

```
49      if(top == -1 ){
50          std::cout << "Stack underflow." << std::endl;
51          return true;
52      }
53      return false;
54  }
55  bool isFull(){
56      //not equal to maxCap
57      if(top >= elements){
58          std::cout << "Stack overflow." << std::endl;
59          return true;
60      }
61      return false;
62  }
63  void push(){
64      // check if the stack is full, if yes return an error
65      if(isFull()) return;
66      // ask the user for a value to be printed in the stack
67      std::cout << "What number you want to add in the stack? ";
68      std::cin >> newData;
69      //store inside the stack??
70      stack[++top] = newData;
71  }
72  void pop(){
73      // check if there are no more elements inside the stack
74      if(isEmpty()) return;
75      //display if the element to be popped:
76      std::cout <<"Popping: " << stack[top] << std::endl;
77      //decrement the top
78      top--;
79  }
80  void Top(){
81      //error checking?
82      if(isEmpty()) return;
83      //display the top element
84      std::cout << "The top element is: " << stack[top] << std::endl;
85  }
86
87  void displayAll(){
88      //error checking?
89      if(isEmpty()) return;
90      //loops backward to the top
91      for(int i = top; i >= 0; i--){
92          std::cout << stack[i];
93      }
94  }
```

Output of the Code:

```
Enter the number of maximum elements for the huge stack (Max is 100): 67
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
Enter a number:
```

```
Enter a number: 1
What number you want to add in the stack? 660
```

```
Enter a number: 2
Popping: 660
```

```
Enter a number: 3
Stack underflow.
```

```
Enter a number: 4
Stack underflow.
1
```

```
Enter a number: 5
0
```

```
Enter a number: 6
Stack underflow.
```

```
Enter a number: 6

660Stack Operations:
```

```
Enter a number: 1
What number you want to add in the stack? 770
```

```
Enter a number: 1
What number you want to add in the stack? 696
```

```
6. Display All
Enter a number: 6

696
770
660Stack Operations:
```

Analysis:

Similar to ILO A, the code uses a prototype to use the methods that we're used. In the function declarations, the code used if/else statements to check if the top is empty, not equal to the max capacity (maxCap), if the stack is full, if the user wants to print in the stack, and error checking. In this displayAll function, we used a for loop so the stack can be displayed from backward to top meaning that once a stack was pushed, it will enter first in the bottom before entering another and it will display at the top of it. Back in the main function, we used cin so we can enter the number of elements and use a while loop to enter the choices so it can be display.

Note: This procedure output is prohibited from using AI by Engr. Jimlord Quejardo.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

N/A

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